

PG-MoE

Phase-Gated Mixture-of-Experts
for Multimodal Human Activity Recognition

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MAS.S60 / 6.S985 · Multimodal AI

RGB Video × IMU Sensor × Physics-Driven Gating

Why Phase-Aware Fusion?

The Problem

Existing multimodal fusion treats all timesteps equally

But modalities have different strengths at different phases of an action

Our Insight

Use biomechanical phase features from IMU to dynamically gate modalities

Physics-driven, not black-box attention

Example: Throw Action

Preparation

Trust: Vision
See posture

Acceleration

Trust: IMU
Acc peak

Impact

Trust: IMU
Motion blur

Follow-through

Trust: Vision
See recovery

Dataset & Experimental Setup

UTD-MHAD Dataset

861 samples, 27 action classes

8 subjects (4 train / 4 test)

4 modalities: RGB, Depth, Skeleton, IMU

We use: RGB + IMU only

(more practical — phones have cameras + IMU)

Why RGB + IMU?

Prior SOTA uses Skeleton+Depth → 98%+

but requires Kinect (specialized hardware)

RGB+IMU = smartphone sensors

→ Deployable, practical, real-world

→ More challenging but more relevant

Modality Complementarity (per-class accuracy)

Action	IMU	Vision	Better
Tennis serve	50%	94%	Vision +44%
Throw	25%	56%	Vision +31%
Boxing	69%	100%	Vision +31%
Pickup & throw	69%	100%	Vision +31%
Baseball swing	81%	100%	Vision +19%
Wave	100%	75%	IMU +25%
Draw circle CW	100%	75%	IMU +25%
Draw circle CCW	100%	75%	IMU + 25%
Arm curl	88%	69%	IMU + 19%
Clap	100%	100%	Both strong
Walk	100%	100%	Both strong
Squat	100%	100%	Both strong

(if perfect gating) **Oracle fusion: 98.4%**

PG-MoE Architecture

Vision Expert

ResNet3D (Kinetics)
fine-tuned on UTD-MHAD
→ (B, T_v, 256)
89.3% standalone

Gated Fusion

$$z(t) = \alpha(t) \cdot v(t) + (1 - \alpha(t)) \cdot i(t)$$

$\alpha=1$ → trust Vision
 $\alpha=0$ → trust IMU

$\alpha(t)$

Classifier
→ 27 classes

IMU Expert

Deep 1D-CNN (ResBlocks)
stride-2 downsampling
→ (B, 12, 256)
82.3% standalone

Phase Arbitrator

Raw IMU → Physics features:
 $|a|, d^2|a|/dt^2, d/dt(|a|^2)$
→ PhaseEncoder (3→32)
→ MLP → σ → $\alpha(t)$
Only 2,881 params!

Key Novelty

Gating driven by
physics features,
not learned attention
→ interpretable

Results

Method	Type	Accuracy
IMU-only (Deep CNN)	Single	82.3%
Vision-only (ResNet3D)	Single	89.3%
<i>Vision-only (Qwen ViT)</i>	Single	58.6%
Early Fusion	Fusion	90.2%
Late Fusion (avg logits)	Fusion	92.3%
Cross-Attn (no gating)	Fusion	89.8%
PG-MoE (fine-tuned)	Ours	91.4%

Finding 1

All fusion methods beat single-modality best (89.3%)

Finding 2

Simple Late Fusion > learned fusion on 431 samples (dataset limitation)

VLM vs Task-Specific Models

Qwen2.5-VL (frozen): 58.6%
ResNet3D (fine-tuned): 89.3%

- VLM encodes semantics, not motion
- Task-specific pretraining matters for HAR

Why PG-MoE Matters

Cross-Attn alone: 89.8% (overfits)
Late Fusion: 92.3% (no interpretability)

- PG-MoE: 91.4% + interpretable $\alpha(t)$
- Know WHEN to trust each modality

Analysis Vision

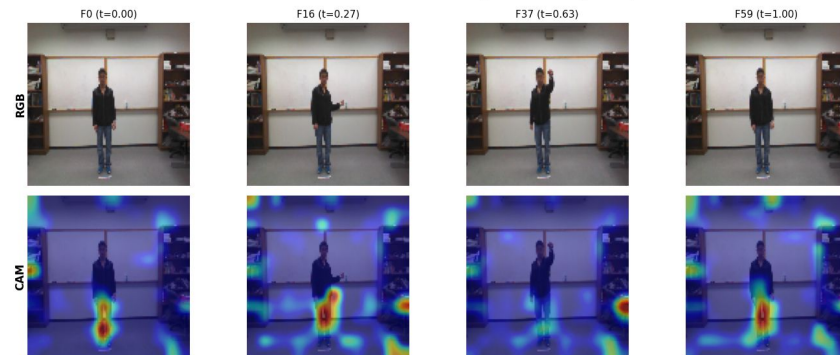
Grad-CAM Analysis

ResNet3D **focuses on person**
but also background edges →
partial scene bias motivates
IMU as complement

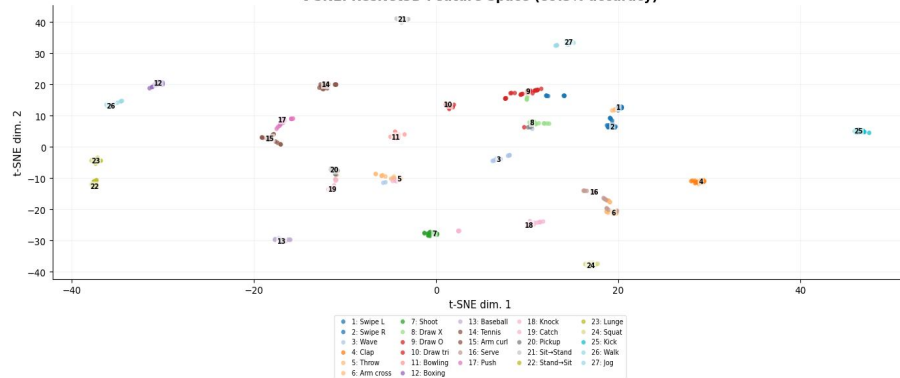
t-SNE Clustering

27 action classes form
well-separated clusters in
ResNet3D feature space
(89.3% accuracy)

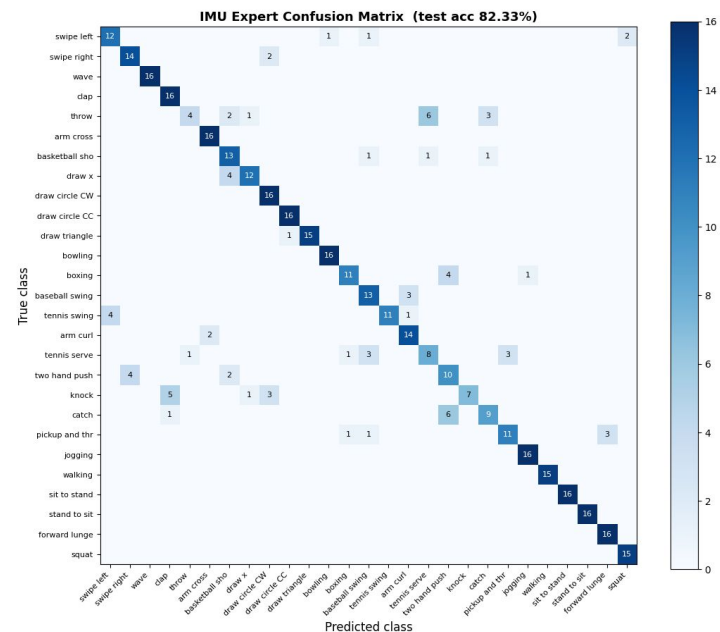
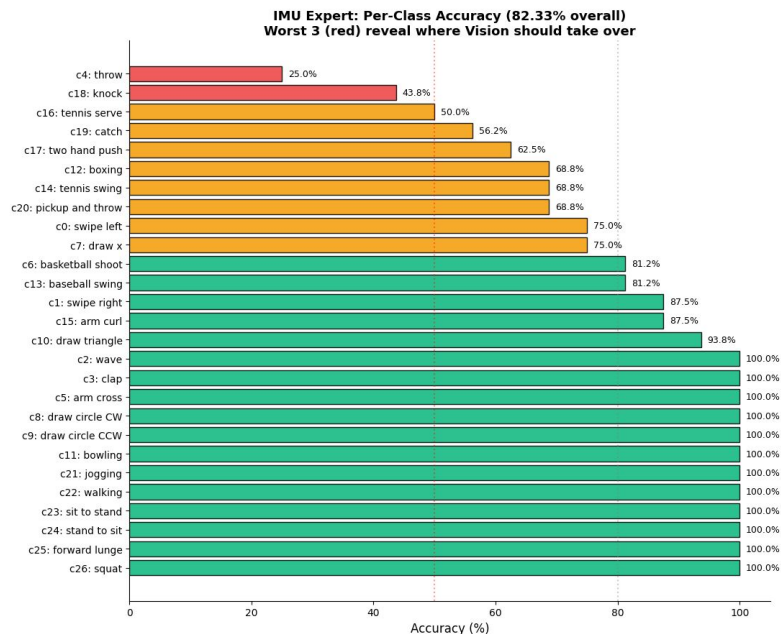
Grad-CAM: ResNet3D attention during throw action (pred=5)



t-SNE: ResNet3D Feature Space (89.3% accuracy)



Analysis: IMU Expert

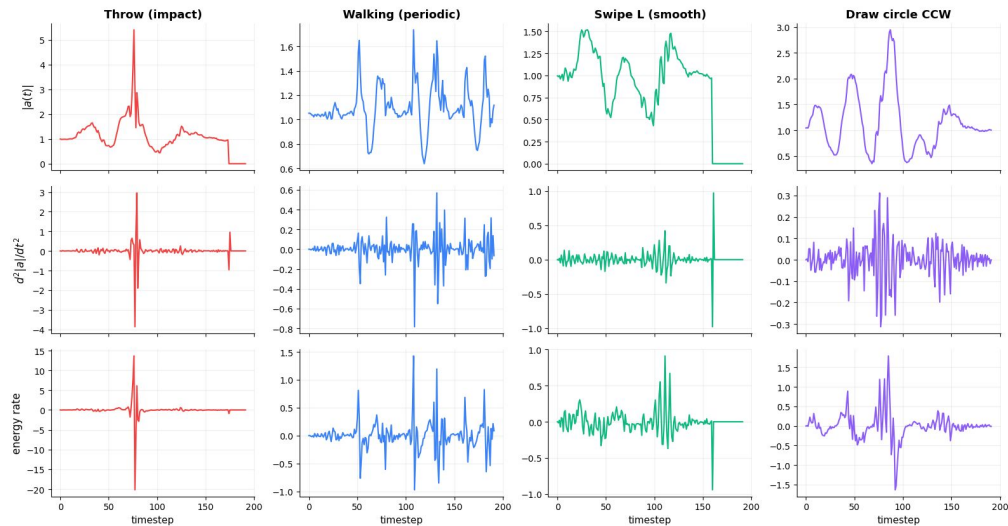


What works/fails

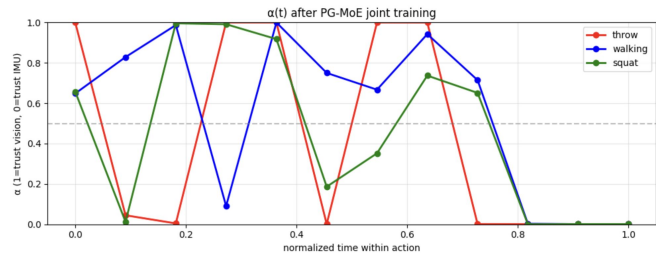
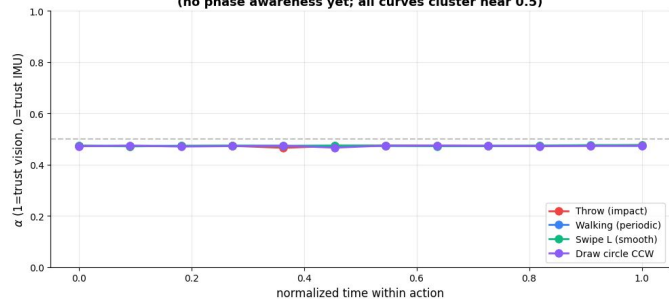
IMU Expert reaches 82.33% (+14.4 pp over midterm), dominating on periodic motions but failing on impact actions — throw 25%, knock 44%, tennis serve 50% — exactly where Vision must take over.

Analysis: Phase-Aware Arbitrator

Phase Features Differ Sharply Across Action Types
(input to Phase Arbitrator)



$\alpha(t)$ — Untrained Baseline
(no phase awareness yet; all curves cluster near 0.5)



How it works & What $\alpha(t)$ learns

Three physical signals ($|a|$, jerk, energy rate) feed a 2.8 K-param MLP producing $\alpha(t)$. After joint training, α diverges sharply per action (throw, walking, squat each take a different path) — phase awareness is learned end-to-end, not hardcoded.

Contributions & Future Work

Contributions

1. PG-MoE framework: physics-driven modality gating for HAR

2. Comprehensive ablation study: 9 methods compared systematically

3. VLM vs task-specific analysis: Qwen ViT 58.6% vs ResNet3D 89.3%

4. Oracle analysis reveals 98.4% upper bound for RGB+IMU fusion

Future Directions

1. Larger datasets (ActionSense, EgoGesture) for learned gating

2. Add Skeleton modality for tri-modal PG-MoE ($\rightarrow 98\%+$)

3. Egocentric HAR with smartwatch IMU + phone camera

4. Contrastive alignment of RGB and IMU feature spaces